



Spatial Spreading of Altruism

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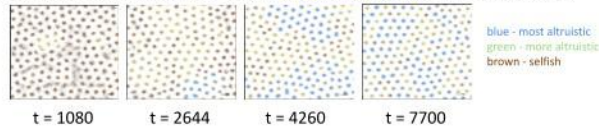
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Motivation

- How can altruistic behaviors evolve and persist when they do not benefit individuals?
- The **group fitness hypothesis** gives one explanation for this apparent paradox
 - Altruistic individuals are at a disadvantage, but altruistic *groups* may outcompete other groups
 - Many theories of grouping: kin selection, similarity of traits, region, etc.
- [Hermesen '22] models altruistic evolution through spatial effects: altruistic individuals benefit individuals nearby, and all individuals compete for resources within a slightly larger radius. They found:

1) spontaneous evolution towards altruism 2) *spatial spreading* of the altruistic trait



Their model was *agent-based*: they simulated discrete individuals. This involves randomness in many places: in 1) movement, 2) reproduction, 3) mutation, and 4) death

Main Question

Does the same spatial spreading of altruism happen, without randomness, in a deterministic, continuous model?

Our Model

- In the place of individual agents, we consider a continuous population distribution $p(x,t,a)$
 - x is in a 1D periodic domain, and a is the altruism trait, ranging from 0 (totally selfish) to 1 (totally altruistic)

Evolution equation:

$$\frac{\partial \rho}{\partial t} = \rho(1 - c \cdot a) + \frac{b_{\max} A(\rho)}{b_0 + A(\rho)}(1 - G_c \cdot \rho) - d\rho + k_D \frac{\partial^2 \rho}{\partial x^2}$$

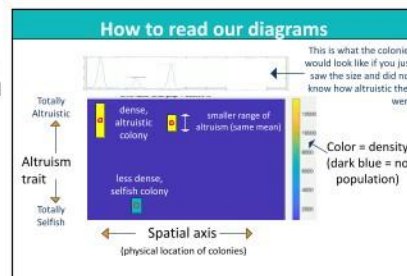
Annotations:
- ρ : population density
- $c \cdot a$: cost of altruism
- $b_{\max} A(\rho) / (b_0 + A(\rho))$: benefit term and nearby altruism
- $(1 - G_c \cdot \rho)$: cost of competition
- $d\rho$: death
- $k_D \frac{\partial^2 \rho}{\partial x^2}$: motility

- The main features of the Hermesen model persist, but are now deterministic:
 - Altruistic population benefits the birth rate of all others nearby
 - Resource competition decreases birth rate
 - Mutation [not shown above] becomes dispersal of the altruism trait – new population has its altruism trait spread out slightly around the level of the parent (equally in both directions)
 - the nonlocal kernel chosen for deterministic mutations is the same exponential distribution as was used for random mutations
 - Spatial movement → diffusion
 - Death is at a uniform rate

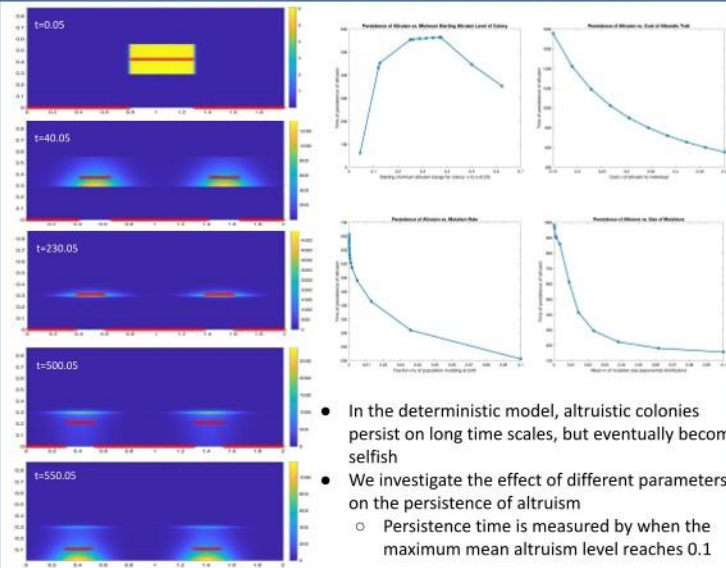
- All parameters have the same mean values as those in the random model

Implementation

We simulate the model in Matlab using finite differences. We use an implicit method for the spatial diffusion term, while the rest of the terms are calculated explicitly at each time step.



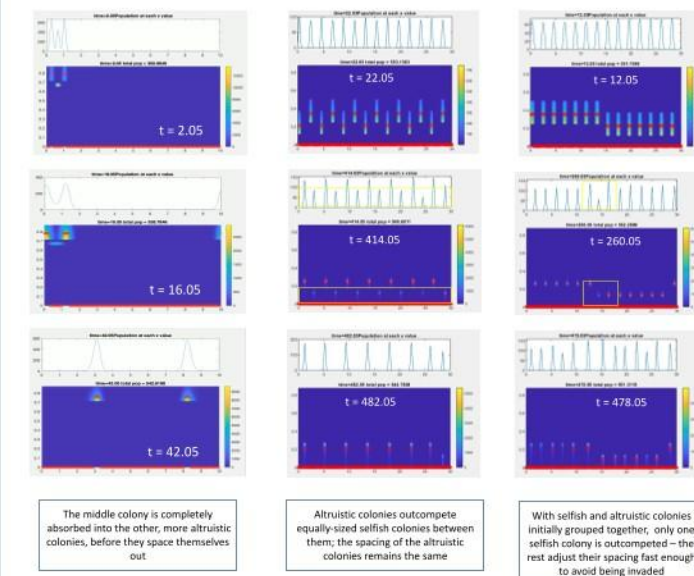
Results: Persistence of Altruistic Colonies



- In the deterministic model, altruistic colonies persist on long time scales, but eventually become selfish
- We investigate the effect of different parameters on the persistence of altruism
 - Persistence time is measured by when the maximum mean altruism level reaches 0.1

Results: Out-competition of Altruistic vs. Selfish Colonies

When spaced closely, we find altruistic colonies out-compete nearby selfish colonies. However, once they reach equilibrium spacing, they coexist.

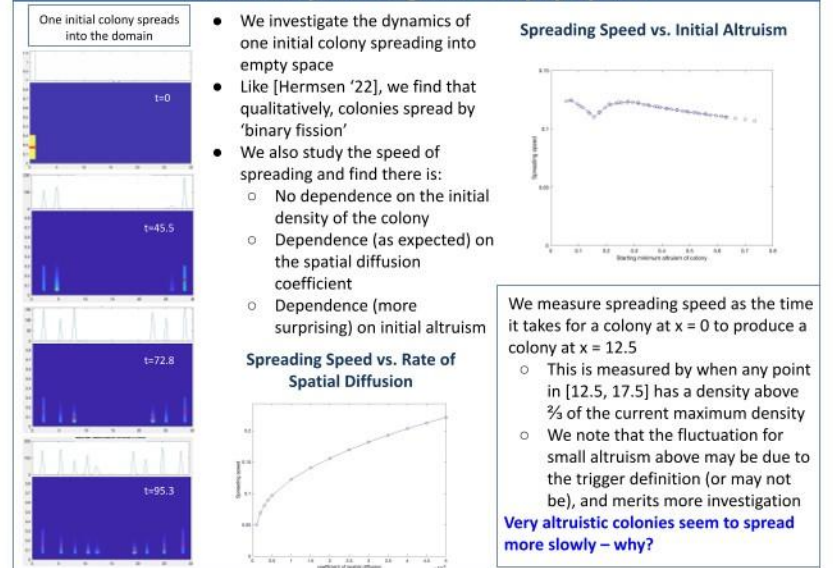


The middle colony is completely absorbed into the other, more altruistic colonies, before they space themselves out

Altruistic colonies outcompete equally-sized selfish colonies between them; the spacing of the altruistic colonies remains the same

With selfish and altruistic colonies initially grouped together, only one selfish colony is outcompeted – the rest adjust their spacing fast enough to avoid being invaded

Results: Spreading into Empty Space



- We investigate the dynamics of one initial colony spreading into empty space
- Like [Hermesen '22], we find that qualitatively, colonies spread by 'binary fission'
- We also study the speed of spreading and find there is:
 - No dependence on the initial density of the colony
 - Dependence (as expected) on the spatial diffusion coefficient
 - Dependence (more surprising) on initial altruism

Spreading Speed vs. Rate of Spatial Diffusion

Conclusions

- The continuous, deterministic model reproduces some, but not all, phenomena seen in [Hermesen '22]:
 - Altruistic colonies outcompete nearby selfish colonies, but only when closely spaced
 - Notably, this implies the persistent, spatial spreading in [Hermesen '22] is **not** seen
 - As in [Hermesen '22], colonies seem to spread into empty space by binary fission
- Altruistic colonies eventually become selfish
 - This did occur for individual colonies in Hermesen's model, and was only mitigated by the spreading of other altruistic colonies.
- When colonies are selfish, structure/grouping decays into a more uniform distribution (predicted by pattern formation theory)
- Persistence of altruistic colonies appears to correlate with initial altruism values, cost of altruism, mutation rate, and magnitude of mutation

Future Questions

Short-term

- Are there other parameters that influence persistence of altruism? (Other system parameters? Presence of selfish colonies? Initial diversity of altruism levels?)
 - States that persist on long but finite time scales are sometimes called "metastable" – can existing theory (e.g. [2]) explain the time scale for persistence of altruism?
- When colonies spread into empty space, the colonies that split off seem to be less altruistic – why?
- Long-term
 - How does the introduction of randomness influence colony formation, colony persistence, and spreading of altruism?
 - Since altruistic colonies out-compete selfish colonies when they are close, would only randomness in motility be enough to see persistent spatial spreading?
- When spreading is present, how can it be described using dynamics?
- How can our parameters and initial conditions relate to or influence real-life phenomena?

References

- Hermesen R. Emergent multilevel selection in a simple spatial model of the evolution of altruism. *PLoS Comput Biol*. 2022 Oct 25;18(10):e1010612. doi: 10.1371/journal.pcbi.1010612. PMID: 36282807; PMCID: PMC9595567.
- Iron, David, and Michael J. Ward. "A Metastable Spike Solution for a Nonlocal Reaction-Diffusion Model." *SIAM Journal on Applied Mathematics*, vol. 60, no. 3, 2000, pp. 778–802.